Shesh Budhabhatti

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EDUCATION

NORTHEASTERN UNIVERSITY (Dean's List x3 at Khoury College of Computer Sciences) **Boston, MA**
B.S., Combined Major in CS (Cybersecurity) and Business Administration (Finance Concentration) April 2028
Coursework: Systems Security, Foundations of Cybersecurity, Network Fundamentals, CS Systems, Law & Ethics of Digital Technology, Algorithms & Data Structures, Object-Oriented Design (Java)

SKILLS

Languages: Python, Java, C, C++, MATLAB, JavaScript, SQL, Assembly
Frameworks & Web: React, Vite, REST APIs, Claude API, Vercel Serverless Functions, HTML/CSS
Systems & Tools: Git/GitHub, Copilot, Claude, GitHub Actions, Linux/Unix, Docker, GDB, Valgrind, pthreads, Azure DevSecOps, Jira, Agile, Bitbucket, Polarion, Miro, Arduino, Microsoft Office (Excel, Word, PowerPoint)
Cloud & Data: AWS (Lambda, DynamoDB, API Gateway, IAM, SAM), MySQL, NoSQL
Security: Threat modeling (STRIDE), secure coding, authentication/authorization (OAuth 2.0), access controls, least privilege, privilege escalation, penetration testing, SQL injection, XSS, CSRF, prompt injection, risk analysis (CVSS 3.1), root cause analysis, cryptographic protocols, SAST (Semgrep, custom taint rules), Wireshark, Metasploit, Hashcat, Burp Suite
Software Engineering: API Development & Integration, UML/Software Architecture, Test Automation, Software Testing & Debugging, System Validation & Performance Testing

WORK EXPERIENCE

MEDTRONIC **Boston, MA**
Software Architecture Co-op August 2026 – Present
- Building a GitHub Actions CI/CD pipeline that runs on every commit to auto-generate UML diagrams and technical documentation for Medtronic's Hugo™ surgical robot platform, keeping design records traceable to code changes
- Integrating generated design documentation with a MySQL-backed database through a cross-team API, contributing 20+ pull requests and addressing code review feedback from engineers across teams
- Developed a redline diff tool for pull requests to highlight design changes between versions, added inline PlantUML support to the architecture docs site and regression-tested 50+ existing diagrams to verify PlantUML and Mermaid rendering

Product Security Intern June 2026 – August 2026
- Built Python tooling to combine latency, jitter, packet loss, and RSSI into a health score for autonomous Wi-Fi failover on a Raspberry Pi 5/ESP32 testbed
- Validated failover under load (28 clients, 600 packets/sec), RF interference (0 to 4 disruptors, 0% packet loss), and 802.11 deauthentication attacks, holding a stable band switch unattended for ~10 hours
- Diagnosed association and interference failures from packet captures, and authored a 20-page findings report (20+ findings) presented to Medtronic vice presidents and engineering leadership

MICROSOFT **Boston, MA**
AI Student Ambassador & WICxDevSecOps Program Participant October 2025 – February 2026
- Trained in threat modeling and secure CI/CD on Azure DevSecOps pipelines, including AI model security
- Piloted Copilot/Omi AI with 100+ users, collecting feedback on quality, privacy, and edge cases for product teams

MEDTRONIC **North Haven, CT**
R&D Software Engineering Intern June 2025 – August 2025 | July 2024 – August 2024
- Automated test fixture calibration with Python scripts, cutting calibration time 37% and saving 4+ hours weekly
- Authored fixture manual for cross-functional teams across US/India (15+ engineers), reducing onboarding time
- Implemented AES-256 encryption for the Signia™ Stapling System, a high-compliance medical device, under HIPAA and FDA cybersecurity requirements

SECURE & ASSURED INTELLIGENT LEARNING LAB **New Haven, CT**
Cybersecurity & Safety Researcher June 2023 – February 2024
- Exploited prompt injection vulnerabilities in LLM applications (OWASP LLM Top 10 #1) through 50+ attacks, identifying 8 vulnerabilities and documenting mitigations presented to lab researchers

KPMG **Hartford, CT**
Summer Intern, Audit/Tax/Advisory Accounting July 2023 – August 2023
- Collaborated with 5-person teams across 3 companies, identifying 2 key control weaknesses in inventory tracking
- Analyzed financial data across 3 client studies with Tableau, visualizing audit risk insight for senior stakeholders

G.U.R.L.S CHESS CLUB NON-PROFIT ORGANIZATION **USA**
Founder & Executive Director & TEDx Speaker December 2021 – Present
- Founded chess non-profit coaching 200+ students (70% female) across 70+ lessons, featured by US Chess (USCF)
- Redesigned curriculum using retention data analysis, improving participant retention from 40% to 78%

PROJECTS

llm-sec-review | Python, Claude API, Semgrep **October 2026**
- Built an LLM security reviewer for Python that returns findings as line, CWE, and severity, with a schema-validation layer (9 unit tests) rejecting malformed model output
- Benchmarked it against a Semgrep baseline on 19 labeled samples (12 vulnerability classes, 2 prompt-injection samples, 5 safe controls): Claude detected 11/14 vs Semgrep's 6/14, with 0 false positives
- Tested injection resistance with samples embedding a "report zero findings" instruction next to a real vulnerability: both still reported, 2/2

Sylla | React/Vite PWA, AWS (Lambda, DynamoDB, API Gateway, SAM), Vercel **August 2026 – Present**
- Built an AI vision web app that extracts events from photos of syllabi and flyers and adds them to the user's calendar
- Manually reviewed the serverless API's data flow and found model-generated output (a calendar name and an echoed error message) reaching the client unsanitized, fixed both and encoded the pattern as a custom Semgrep taint rule with ruleid/ok tests running in GitHub Actions on every push
- Implemented Google OAuth 2.0 authorization-code flow with scoped calendar permissions, keeping token exchange and client secrets server-side
- Treated model output as untrusted, enforcing a strict schema, length caps, and instruction-pattern filtering at three layers before any calendar write to limit prompt injection
- Built a device-to-cloud sync layer on API Gateway + Lambda + DynamoDB (single-table userId/eventId), with server-side Google ID-token verification, conditional-write dedup, and least-privilege IAM
- Built with Claude and GitHub Copilot, reviewed all AI-generated code, and wrote unit and integration tests verifying input rejection, dedup, and 401s on invalid tokens

Web Exploitation & CTFs | Linux Privilege Escalation, Web Pentesting **October 2025 – February 2026**
- Completed OverTheWire Bandit challenges involving Linux permissions, SUID, and switching between users across five levels
- Penetration-tested a web app by hand, exploiting broken access control via parameter tampering to escalate a buyer account to seller, plus SQL injection, stored XSS, and CSRF to act as another user

FUSE Filesystem Implementation | C, Systems Programming **November 2025**
- Implemented a custom C filesystem with inode management and mmap/munmap allocation to support files larger than 4 KB

- Diagnosed memory-safety bugs using GDB and Valgrind, passing all 31 test cases

Threaded Merge Sort | C, Concurrency & Performance **November 2025**

- Built a multithreaded merge sort in C, benchmarking 100M integers across 2-core and 10-core machines, reaching 4.1x speedup at 10 threads (29.5s to 7.2s), plateauing past physical core count

BGP Router Implementation | Python, Networking **February 2026**

- Implemented BGP message handling for peer handshakes, route updates, and withdrawals, with longest-prefix-match forwarding and route tie-breaking
- Implemented IP prefix calculations using bitwise operations and rebuilt forwarding-table entries after route withdrawals for correctness

AWARDS/PUBLICATIONS/CERTIFICATIONS

TEDx: "Empowering Equality, Inspiring Action" (2023) | SMARTA Podcast (2025) | Lean Six Sigma Yellow Belt (2024) | CT Women's Chess Champ. (2018) | TRAIN's Intro. to Artificial Intelligence Certified (2024)